

GCSE algebra - Set 1

TMUA practice questions by topic. Specification M4.1-M4.8, M4.18, M4.19. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Make x the subject of $y = \frac{2x+3}{x-1}$.
- A** $x = \frac{3-y}{y-2}$
B $x = \frac{y-2}{y \pm 3}$
C $x = \frac{y \pm 3}{y \mp 3}$
D $x = \frac{y \mp 3}{y \mp 3}$
E $x = \frac{y \mp 3}{2y-3}$
F $x = \frac{2y-3}{y-1}$
- 2 Find the n th term of the quadratic sequence 3, 8, 15, 24, 35, ...
- A** $n^2 + 2n$
B $(n+1)^2 + 1$
C $n^2 + 3n - 1$
D $n^2 + 2$
E $n^2 + n + 1$
F $2n^2 + 1$
- 3 Which of the following is an identity (true for all values of x)?
- A** $(x+3)^2 - (x-3)^2 = 12x$
B $x^2 + 6x + 9 = 0$
C $(x+3)^2 = x^2 + 9$
D $(x-3)^2 = x^2 - 6x - 9$
E $(x+3)(x-3) = x^2 + 9$

- 4 A car accelerates uniformly from rest to 20 m/s in 8 seconds, travels at 20 m/s for 12 seconds, then decelerates uniformly to rest in 10 seconds. Find the total distance travelled.

A 600 m
 B 400 m
 C 420 m
 D 300 m
 E 380 m
 F 480 m

- 5 Given that $v^2 = u^2 + 2as$, find s when $v = 13$, $u = 5$ and $a = 4$.

A 18
 B 21
 C 36
 D 12
 E 9
 F 24

- 6 Simplify $\frac{3}{x+2} - \frac{2}{x-1}$.

A $\frac{5x-1}{(x+2)(x-1)}$
 B $\frac{x+4}{(x+2)(x-1)}$
 C $\frac{x+3}{x-7}$
 D $\frac{(x+2)(x-1)}{(x+2)(x-1)}$
 E $\frac{(x+2)(x-1)}{(x+2)(x-1)}$
 F $\frac{(x+2)(x-1)}{(x+2)(x-1)}$

- 7 Given that $T = 2\pi\sqrt{\frac{l}{g}}$, express l in terms of T and g .

A $l = \frac{gT}{2\pi^2}$
 B $l = \frac{4\pi^2g}{T^2}$
 C $l = \frac{4\pi^2g}{T^2}$
 D $l = \frac{4\pi^2g}{T^2}$
 E $l = \frac{4\pi^2g}{T^2}$
 F $l = 4\pi^2gT^2$

- 8** A ball is thrown vertically upwards and its height after t seconds is $h = 20t - 5t^2$ metres. For how long is the ball at least 15 m above the ground?
- A** 4 s
B 1 s
C 0.5 s
D 1.5 s
E 2 s
F 3 s
- 9** Expand and simplify $(x + 2)(x - 1)(x + 3)$.
- A** $x^3 + 4x^2 + 7x - 6$
B $x^3 + 2x^2 - 5x - 6$
C $x^3 - 4x^2 + x + 6$
D $x^3 + 4x^2 - x - 6$
E $x^3 + 6x^2 + 11x + 6$
F $x^3 + 4x^2 + x - 6$
- 10** A sequence has first term 2 and each subsequent term is obtained by squaring the previous term and subtracting 2. What is the 50th term?
- A** 2^{50}
B -2
C 14
D 4
E 0
F 2

11 Factorise $6x^2 - 7x - 20$.

A $(3x - 4)(2x + 5)$

B $(6x - 5)(x + 4)$

C $(3x + 4)(2x - 5)$

D $(3x - 5)(2x + 4)$

E $(6x + 5)(x - 4)$

F $(3x + 5)(2x - 4)$

12 The sum of two numbers is 12 and the difference of their squares is 48. Find the larger number.

A 6

B 10

C 8

D 7

E 4

F 9